

JECRC University, Jaipur
B.Tech.-I Sem I Year 2025-26
Tutorial Sheet-7 (CO3)
Subject: Mathematics-1
Code: DMA024A

CO3: The tool of power series and Fourier series for learning advanced Engineering Mathematics.

Part-A

(2x5=10)

A1. Define sequence with example.

A2. Define monotonic sequence.

A3. Test the convergence of the sequence $\left\{ \frac{n^2+n}{2n^2-n} \right\}$.

A4. Test the convergence of the series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty$.

A5. Define the comparison test.

Part

-B

(3x7=21)

B1. Test the convergence of the series- $\sum_{n=1}^{\infty} \frac{\sqrt[3]{3n^2+1}}{\sqrt[4]{4n^3+2n+7}}$.

B2. Test the convergence of the series - $1 + \frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \dots$.

B3. Test the convergent of the series $\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \frac{5}{4^p} \dots$

Part -C

(3x11=33)

C1. Test the convergent of the series: $\sum_{n=1}^{\infty} [\sqrt{n^4+1} - \sqrt{n^4-1}]$.

C2. Test the convergent of the series: $1 + \frac{1.2}{1.3} + \frac{1.2.3}{1.3.5} + \frac{1.2.3.4}{1.3.5.7} + \dots \infty$

C3. Test the convergent of the series: $1 + \frac{3}{2}x + \frac{5}{9}x^2 + \frac{7}{28}x^3 + \frac{9}{65}x^4 + \dots \infty$

Answers:

A1. A2.	A3. Convergent	A4. Convergent	A5.	B1. Divergent
B2. Divergent	B3. Divergent if $p \leq 2$, convergent if $p > 2$.	C1. Convergent	C2. Convergent	C3. Divergent if $x \geq 1$, convergent if $x < 1$.